Analyzing Business Performance Using Demand Forecasting Techniques

Final Project Report

Submitted by

Yash Yadav

In partial fulfillment of the summer project by

IITK Consulting Group

Academic Year 2022-23

Contents

1 Introduction

The retail industry generates large volumes of transactional data that can be used to understand customer behaviour, optimize operations, and improve profitability. Among the analytical tasks this data enables, *demand forecasting* – estimating future customer demand from historical sales – is one of the most valuable.

1.1 Motivation

Inventory mismanagement is a major source of retail inefficiency. When demand is *underestimated*, products run out, customers are lost, and revenue opportunities disappear. When demand is *overestimated*, excess stock accumulates, storage and wastage costs rise, and working capital is locked in unsold goods. Accurate forecasting therefore has a direct effect on both profitability and customer satisfaction. Retail demand is shaped by seasonality, holidays, regional demographics, and category-specific buying cycles, which makes forecasting challenging but high-value.

1.2 Problem Statement

Four companies operating across seven cities provide over two years of weekly sales, with a holiday flag attached to each week. The dataset poses several challenges: hidden department identities, missing weekly observations, strong seasonality, holiday-driven spikes, and multiple companies and regions. The goal is to forecast demand and, equally important, to explain business behaviour and recommend improvements.

1.3 Objectives

1. Clean the data and reconstruct missing observations.
2. Analyze company, city, holiday, and departmental sales behaviour.
3. Infer the unlabelled department identities from sales patterns.
4. Evaluate multiple forecasting techniques and select the best.
5. Generate a six-month demand forecast.
6. Translate the findings into business recommendations.

1.4 Workflow

The pipeline proceeds as: data understanding $\rightarrow$ cleaning $\rightarrow$ exploratory analysis $\rightarrow$ department inference $\rightarrow$ time-series analysis $\rightarrow$ model development and evaluation $\rightarrow$ forecasting $\rightarrow$ recommendations. The remaining sections follow this order.

2 Dataset Description

The data is a balanced weekly panel of company × store × department observations spanning 2010–2012, recorded for week-ending Fridays. Each company operates seven stores (one per city) and each store contains six departments, allowing direct comparison across companies and regions.

Table 1: Dataset features.

Feature	Description
`Date`	Weekly observation date (week-ending Friday).
`Store`	Store identifier (1–7), mapped to a city.
`Dept`	Department identifier (1–6); names hidden.
`Weekly_Sales`	Weekly sales revenue.
`IsHoliday`	Boolean holiday-week indicator.
`Company`	Retail company name.

A `City` column is derived from the store identifier (Table 2). Department names are deliberately withheld; a key task (Section 5) is to infer them from sales behaviour – for example, a department whose sales rise in winter is likely clothing.

Table 2: Store-to-city mapping used in the analysis.

Store	1	2	3	4	5	6	7
City	Delhi	Mumbai	Kolkata	Lucknow	Bangalore	Indore	Jaipur

3 Data Cleaning and Preprocessing

Forecasting models are sensitive to data quality, so each company dataset was first inspected for shape, data types, missing values, and store/department coverage. Two structural anomalies were found and fixed.

3.1 Spar : Excess Departments

Store 7 in the Spar dataset contained 77 departments rather than the expected six. To keep the panel consistent across all companies, only departments 1–6 were retained, removing the spurious records.

3.2 Seven Eleven : Missing Weeks

In Seven Eleven, missingness appeared not as null cells but as *absent weekly rows*. Re-indexing the data onto the complete store × department × week grid exposed 167 missing records, and these were strongly non-random (Table 3). The single largest block – 124 consecutive weeks in Store 3 (Kolkata), Department 6 (Furniture) – is a contiguous reporting outage rather than genuine zero-sales weeks. Treating it as zero would bias the series downward, so it was reconstructed instead.

Table 3: Missing weekly records in Seven Eleven (total 167).

Store (City)	Dept	Missing weeks
3 (Kolkata)	5	30
3 (Kolkata)	6	124
4 (Lucknow)	6	8
5 (Bangalore)	6	5

3.3 Reconstruction and Validation

Within each `(Store, Dept)` group, the missing weeks were created on the full grid, sorted chronologically, and filled with a three-step hybrid: linear interpolation between known values, then forward-fill (`ffill`) and backward-fill (`bfill`) for the leading and trailing edges that interpolation cannot reach (interpolation needs a known value on both sides). Performing the fill per group is essential to prevent one department's values leaking into another. After reconstruction, every company had a complete weekly series and zero missing `Weekly_Sales`.

4 Exploratory Data Analysis

4.1 Company Performance

Aggregating sales by company gives a clear ordering: D-Mart leads (≈158M), followed by Spar (≈140M) and Bigg Bazar (≈135M), with Seven Eleven well behind (≈89M) (Figure 1).

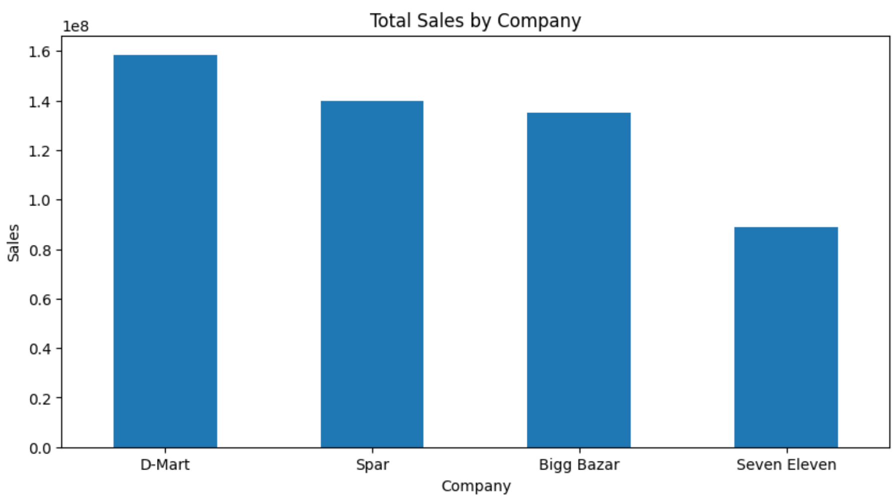

Figure 1: Total sales by company. D-Mart leads; Seven Eleven is the lowest by a wide margin.

Table 4: Approximate total sales by company over the full period.

Company	Total sales (approx.)	Share
D-Mart	158M	30%
Spar	140M	27%
Bigg Bazar	135M	26%
Seven Eleven	89M	17%

The weekly trends (Figure 2) show that the four companies move *together*: they share synchronized peaks around the year-end festive period and differ mainly in scale, not in shape. This indicates demand is driven largely by shared calendar and seasonal factors rather than by company-specific dynamics.

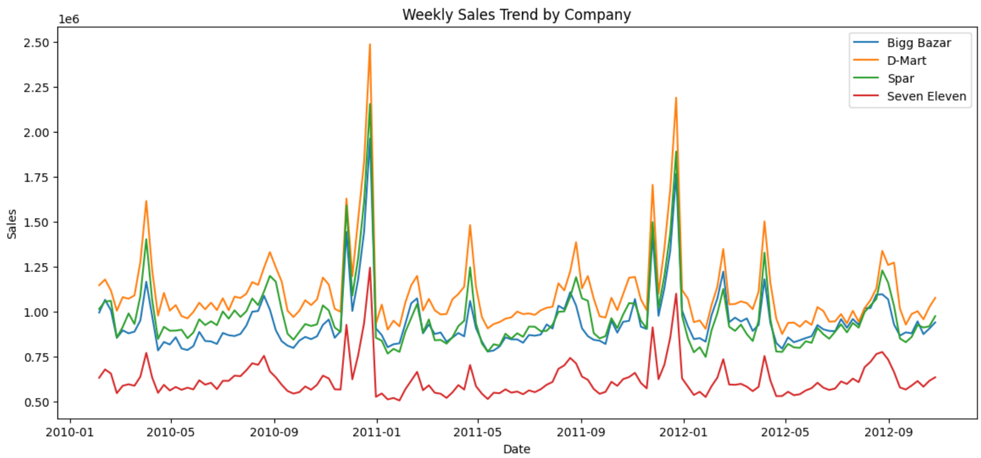

Figure 2: Weekly sales trend by company. The series are synchronized, with pronounced year-end spikes; D-Mart is consistently highest and Seven Eleven lowest.

4.2 City and Regional Behaviour

By city, Mumbai is the strongest market, followed by Indore and Delhi, while Bangalore is the weakest (Figure 3).

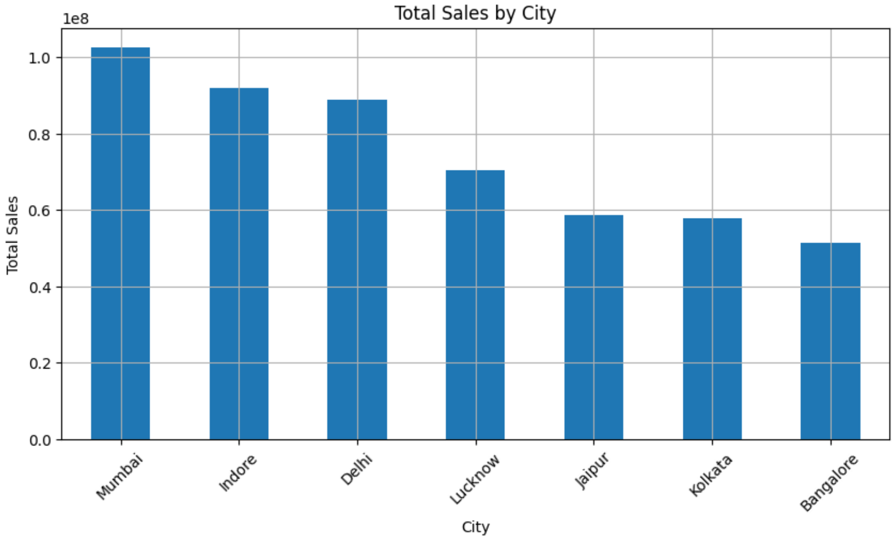

Figure 3: Total sales by city. Mumbai leads and Bangalore trails.

A more useful insight emerges when company and city are crossed (Figure 4): market leadership is *regional*. No single company wins everywhere. Spar dominates Bangalore and Delhi, Bigg Bazar dominates Lucknow, and D-Mart leads Mumbai, Kolkata and Jaipur. Most notably, Seven Eleven – the weakest company overall – is the *leader* in Indore, which proves it possesses a store model capable of winning when executed well.

Figure 4: Company performance across cities. Leadership varies sharply by region; Seven Eleven leads in Indore despite trailing overall.

The regional leaders are summarised in Table 5. The practical implication is that a single national strategy is suboptimal: each company already has regional strongholds, and the largest gains come from closing the gap in the cities where a company trails its own best markets.

Table 5: Leading company by city (from Figure 4).

City	Leading company
Mumbai	D-Mart
Indore	Seven Eleven
Delhi	Spar
Lucknow	Bigg Bazar
Bangalore	Spar
Kolkata	D-Mart
Jaipur	D-Mart

4.3 Seasonality

A classical additive decomposition of the total weekly series (Figure 5) separates trend, seasonal, and residual components. The trend is weak, the seasonal component is strong and dominated by a sharp year-end (November–January) peak, and the residual is small relative to the seasonal swing. This is precisely the structure a *seasonal* forecasting model should be able to exploit, and it warns that a purely non-seasonal model will miss the dominant signal.

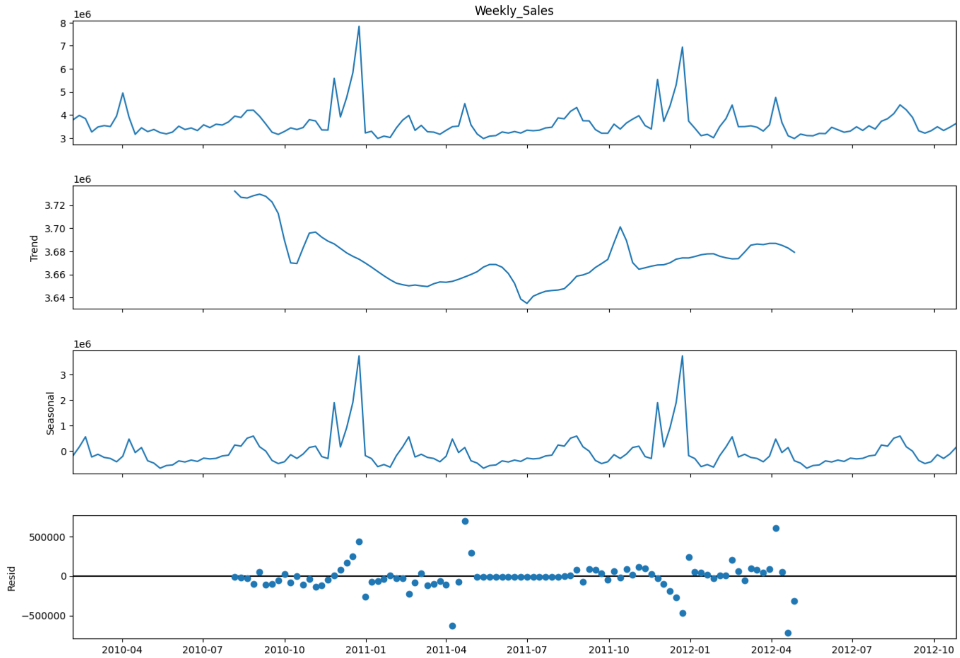

Figure 5: Additive decomposition of total weekly sales: weak trend, strong recurring year-end seasonality, and a small residual.

4.4 Holiday Impact

Comparing mean weekly sales on holiday versus non-holiday weeks shows that every company sells more during holiday weeks (Figure 6).

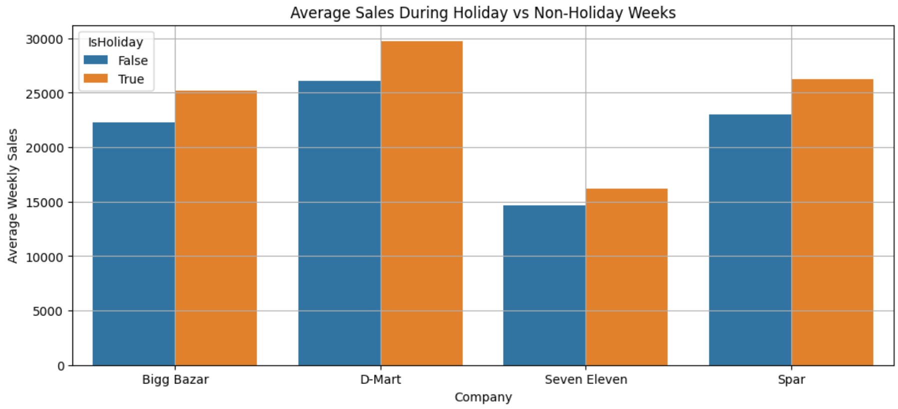

Figure 6: Average weekly sales during holiday vs non-holiday weeks, by company. The holiday uplift is positive for all four companies.

At the department level, however, the effect is highly uneven (Figure 7, Table 6). Electronics (+57%) and Furniture (+75%) surge on holiday weeks, whereas Groceries is flat and even dips slightly – the classic behaviour of an inelastic staple whose demand is steady regardless of festivals.

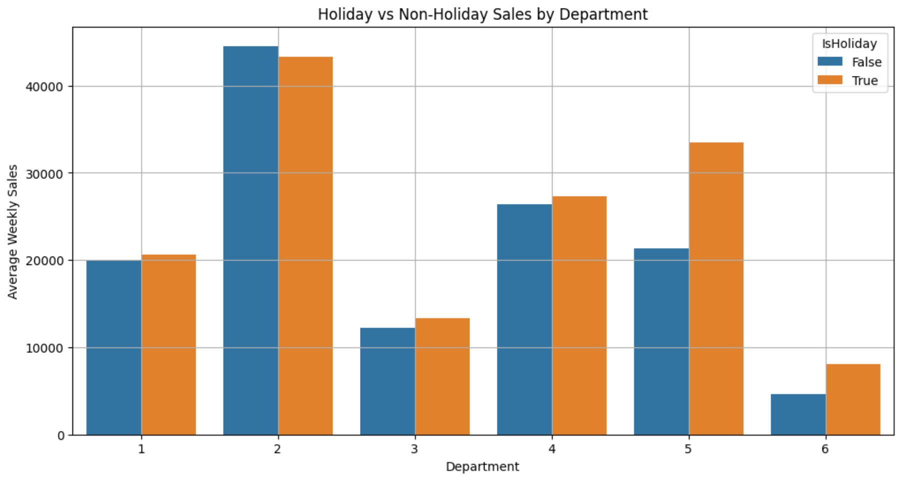

Figure 7: Holiday vs non-holiday mean weekly sales by department. Departments 5 (Electronics) and 6 (Furniture) are strongly holiday-driven.

Table 6: Mean weekly sales: holiday vs non-holiday, by department.

Dept	Inferred category	Non-holiday	Holiday	Lift
1	Clothing & Apparel	20,002	20,677	+3.4%
2	Groceries	44,553	43,270	−2.9%
3	Books & Stationery	12,248	13,362	+9.1%
4	Beverages	26,415	27,275	+3.3%
5	Electronics	21,306	33,522	+**57.3**%
6	Furniture	4,604	8,067	+**75.2**%

4.5 Department Behaviour

Each department has a distinct demand fingerprint (Figures 8 and 9). Groceries (Dept 2) is the largest and almost flat year-round; Electronics (Dept 5) is moderate but surges in November–December; Books & Stationery (Dept 3) has a sharp single peak in August; Furniture (Dept 6) is the smallest and spikes around festivals; Clothing (Dept 1) shows repeated seasonal peaks; and Beverages (Dept 4) is stable and mid-range.

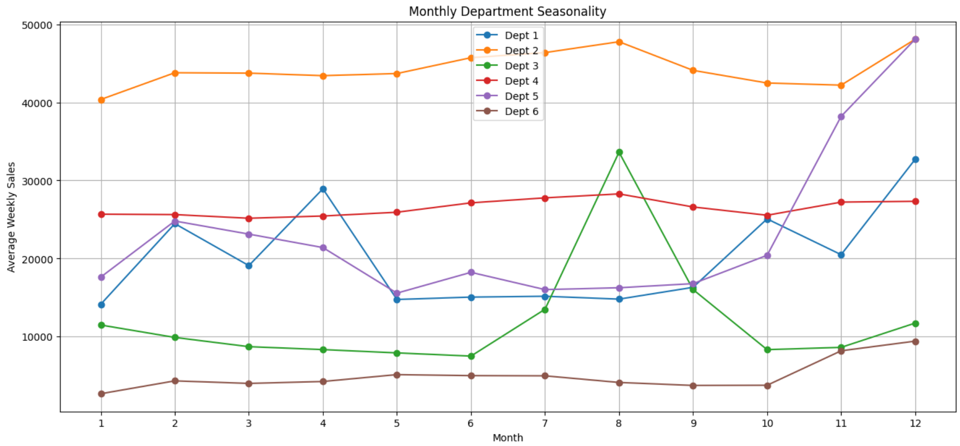

Figure 8: Monthly department seasonality. Note the year-end surge in Electronics (Dept 5) and the August peak in Books & Stationery (Dept 3).

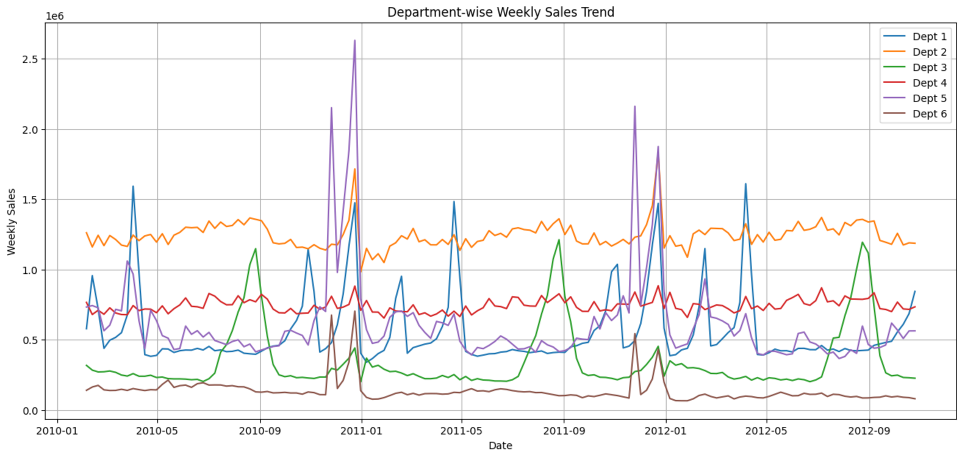

Figure 9: Department-wise weekly sales trend. Electronics produces the largest year-end spikes; Books & Stationery the distinct August peak.

5 Hidden Department Inference

The department identities were inferred from three observable signals – sales *magnitude*, *seasonality* (when demand peaks), and *holiday sensitivity* – and mapped onto the assignment's category list. The mapping is summarised in Table 7; the reasoning follows.

Table 7: Inferred department identities and supporting evidence.

Dept	Identity	Pattern	Reasoning
2	Groceries	Largest, flat, −3% holiday	Inelastic staple bought year-round.
5	Electronics	Strong Nov–Dec, +57% holiday	Big-ticket festive purchases.
6	Furniture	Smallest, +75% holiday, festive spike	Infrequent, sale-event-driven big-ticket.
1	Clothing & Apparel	Repeated seasonal peaks	Wardrobe transitions and festive buying.
3	Books & Stationery	Sharp August peak	Academic-session (back-to-school) demand.
4	Beverages	Stable, mid-range, low sensitivity	Steady routine consumption.

- **Groceries (Dept 2).** The highest and most uniform demand, with a slight *drop* on holidays – the signature of everyday essentials bought continuously and largely indifferent to festivals.
- **Electronics (Dept 5).** A moderate baseline that surges in the festive quarter and lifts by more than half on holiday weeks, matching big-ticket, festival-timed purchases (year-end and Diwali sales events).
- **Furniture (Dept 6).** The lowest baseline but the largest holiday lift (+75%) with a year-end spike – consistent with infrequent, high-value purchases concentrated around sale events.
- **Clothing & Apparel (Dept 1).** Repeated, evenly-spaced peaks align with seasonal wardrobe transitions and festive clothing buying.
- **Books & Stationery (Dept 3).** A single sharp August peak coincides with the start of the academic year.
- **Beverages (Dept 4).** Stable, mid-range, weakly holiday-sensitive demand typical of routine consumption.

The strongest inferences are Groceries, Electronics, Books, and Furniture, whose signatures are unambiguous; Clothing and Beverages are the lower-confidence pair, distinguished mainly by Clothing's seasonal peaks against Beverages' flat profile.

6 Time-Series Analysis and Forecasting Methodology

For forecasting, sales were aggregated into a single weekly total-sales series at weekly (`W-FRI`) frequency. Stationarity was assessed with an Augmented Dickey–Fuller test, and together with the decomposition this informed the differencing and seasonal handling used by the classical models. The series was split chronologically, with the first 80% used for training and the final 20% held out for testing.

6.1 Models

Moving Average forecasts the next value as the mean of the last k observations, a flat smoothing baseline with no trend or seasonality:

$$\hat{y}_{t+1} = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i}. \tag{1}$$

Exponential Smoothing (Holt–Winters, additive seasonality with period $m = 52$) maintains level, trend, and seasonal terms:

$$\ell_t = \alpha\,(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \tag{2}$$

$$b_t = \beta\,(\ell_t - \ell_{t-1}) + (1-\beta)\,b_{t-1}, \tag{3}$$

$$s_t = \gamma\,(y_t - \ell_t) + (1-\gamma)\,s_{t-m}. \tag{4}$$

ARIMA$(2, 0, 2)$ combines autoregression and a moving-average term with backshift operator B:

$$\phi(B)\,(1-B)^d\,y_t = \theta(B)\,\varepsilon_t, \tag{5}$$

capturing trend and short-range correlation but no seasonality.

SARIMA$(1, 0, 1)(0, 0, 1)_{52}$ adds a seasonal block of period $m = 52$ and was fitted on $\log(1 + \text{sales})$ to stabilise variance:

$$\phi(B)\,\Phi(B^m)\,(1-B)^d(1-B^m)^D\,y_t = \theta(B)\,\Theta(B^m)\,\varepsilon_t. \tag{6}$$

Prophet decomposes the series additively into trend and yearly seasonality:

$$y(t) = g(t) + s(t) + \varepsilon_t, \tag{7}$$

where $g(t)$ is a piecewise trend and $s(t)$ is a Fourier-series seasonal term. Prophet handles seasonality natively and is robust to missing data.

6.2 Evaluation Metrics

Models were compared with Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|. \tag{8}$$

RMSE penalises large errors more heavily; MAPE is scale-free and supports direct cross-model comparison.

7 Results and Model Evaluation

Ranked by MAPE (Table 8): Prophet (4.80%), Exponential Smoothing (7.13%), Moving Average (9.69%), ARIMA (11.18%), and SARIMA (32.58%). Prophet is the clear winner.

Table 8: Forecasting model performance (lower is better).

Model	MAE	RMSE	MAPE
Moving Average	330,993.92	381,027.96	9.69%
Exponential Smoothing	237,834.59	360,249.29	7.13%
ARIMA	379,929.25	423,525.98	11.18%
SARIMA	1,095,978.69	1,147,631.43	32.58%
Prophet	**161,658.64**	**236,881.28**	**4.80%**

The most striking result is that the most heavily-parameterised model, SARIMA, is the *worst* working model – beaten even by the naive moving average. Its forecast sits persistently above the actual test values (Figure 12): it systematically over-predicts despite the log transform, and an earlier untransformed attempt diverged completely (MAPE in the hundreds of percent). This is a useful reminder that added complexity carries no guarantee of accuracy – a misspecified seasonal model can underperform a one-line baseline.

Prophet wins because its additive structure – trend plus yearly seasonality – matches the data-generating story uncovered in EDA, and it tolerates the imputed observations gracefully (Figure 13). The simple smoothing baselines were competitive, while ARIMA, lacking a seasonal term, could only track the mean (Figure 11).

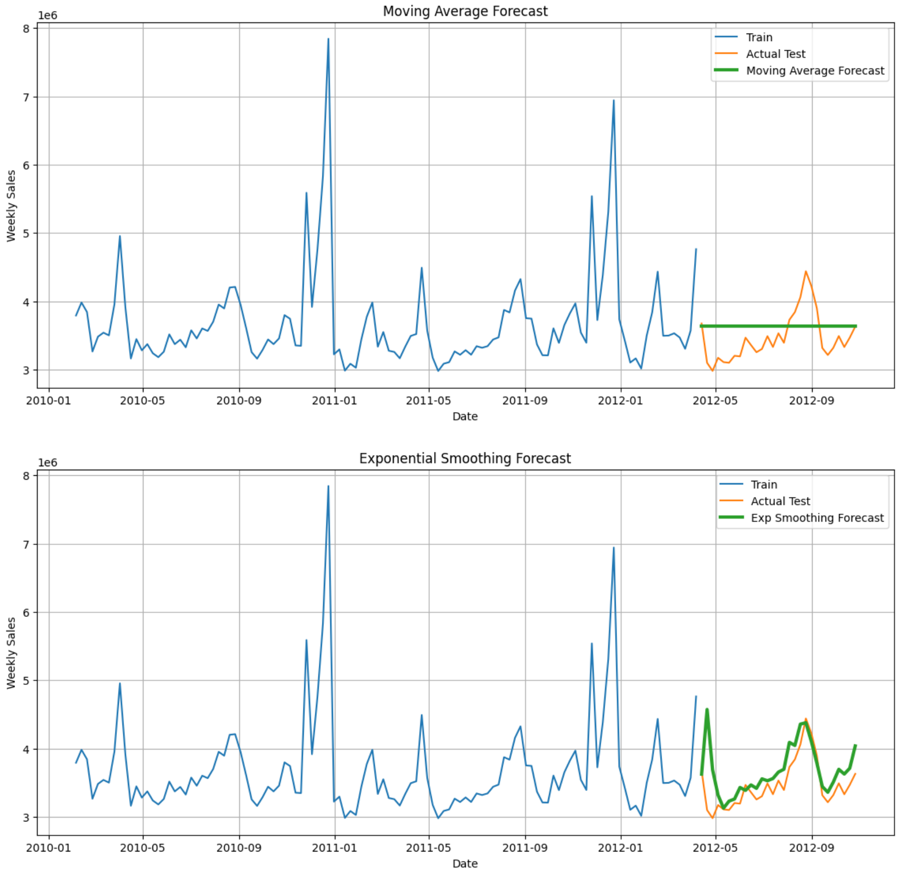

Figure 10: Baseline forecasts: Moving Average (top, flat) and Exponential Smoothing (bottom, captures seasonality).

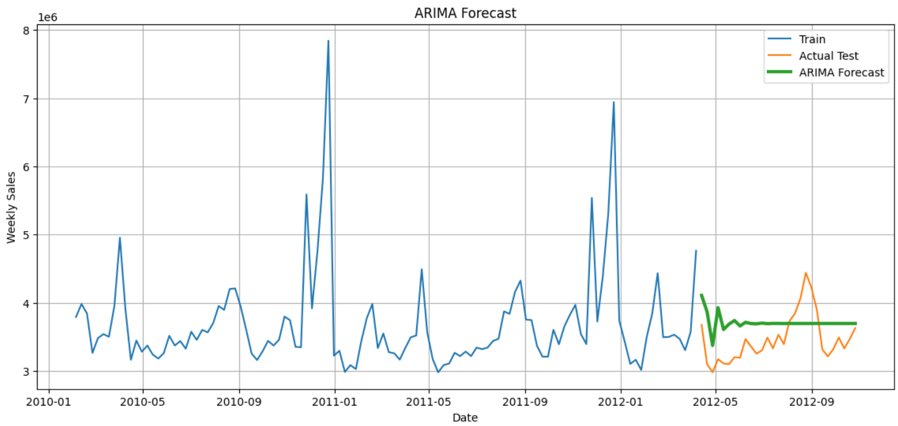

Figure 11: ARIMA(2, 0, 2) forecast – nearly flat, as it has no seasonal term.

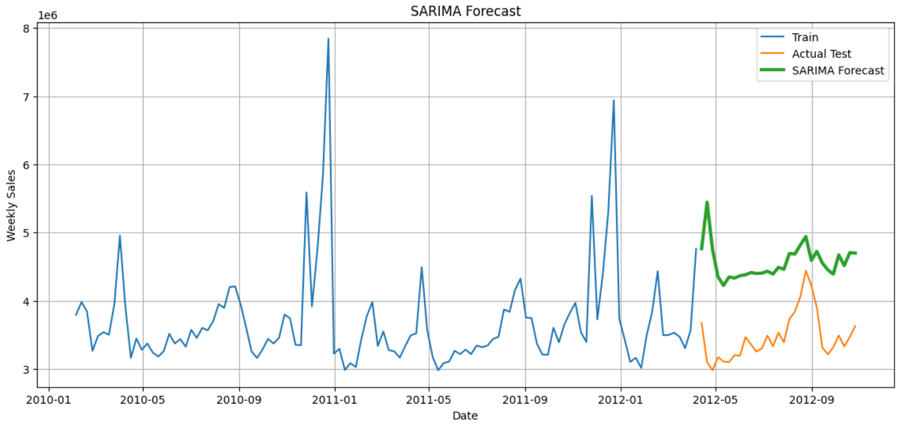

Figure 12: SARIMA forecast. The green forecast sits well above the actual test values, i.e. systematic over-prediction.

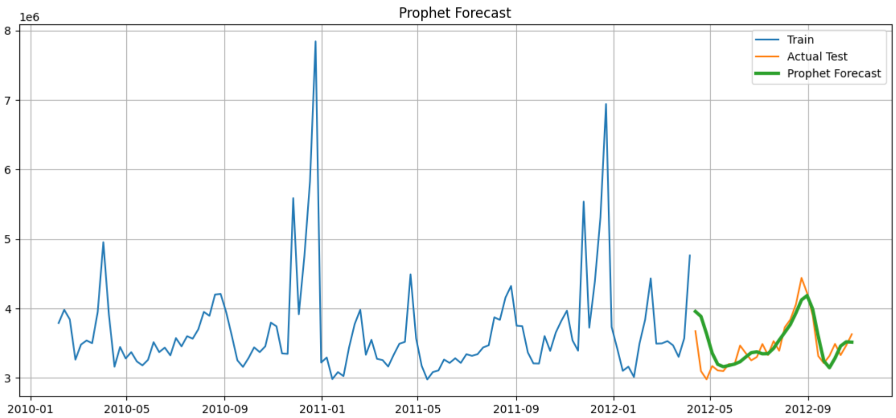

Figure 13: Prophet forecast. It tracks the held-out test period closely, including the dip and recovery.

8 Six-Month Forecast

The retrained Prophet model (fitted on the full series) forecasts 26 weeks ahead with an 80% uncertainty band (Figure 14, Table 9). Weekly totals stay in roughly the 3.5–4.0M range, peaking in late November in line with the festive season, before easing into spring. The uncertainty band widens around the peak, where demand is most volatile – exactly where larger safety stock is justified.

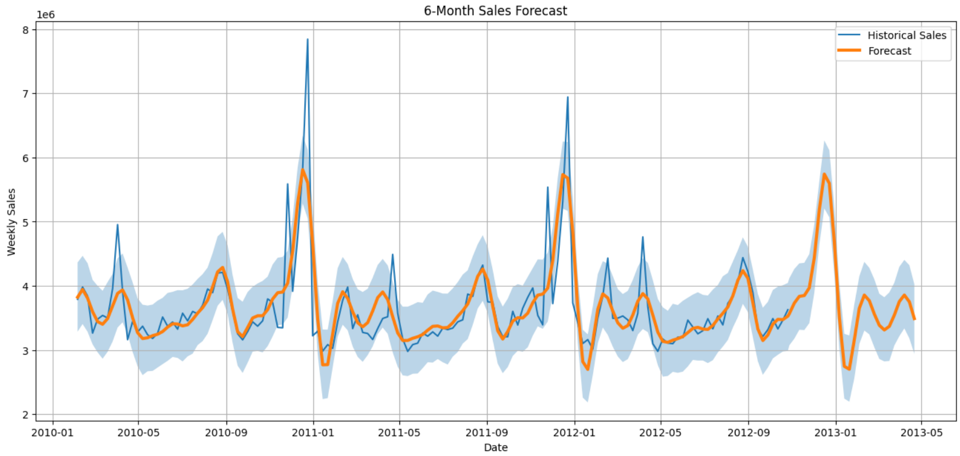

Figure 14: Prophet six-month forecast on the full history with 80% uncertainty interval. The band widens around the seasonal peak.

Table 9: Sample of the Prophet forecast (weekly totals).

Week	Forecast	Lower	Upper
2012-10-28	3,565,162	3,072,881	4,116,842
2012-11-11	3,836,469	3,315,610	4,349,961
2012-11-25	3,969,112	3,419,661	4,494,163
2013-03-31	3,755,195	3,216,307	4,296,695
2013-04-21	3,489,820	2,940,844	4,024,163

9 Business Recommendations

1. Holiday inventory planning. Demand rises sharply before the year-end across every company, concentrated in Electronics (+57%) and Furniture (+75%). Ramp stock and staffing for these categories ahead of the peak and schedule promotions *before* major festivals.

2. City strategy. Mumbai is the strongest market and Bangalore the weakest. Weight inventory toward Mumbai, and treat Bangalore as a recoverable growth gap – audit its marketing and customer engagement and adopt the practices that work in stronger cities.

3. Department management. Stock Electronics aggressively around festivals; protect Groceries *availability* rather than discounting it (promotions add little incremental demand); pre-position Clothing ahead of its seasonal peaks; and run Books & Stationery campaigns in July–August to capture the academic spike.

4. Lift the weak performer (Seven Eleven). Seven Eleven is the lowest company overall yet *leads in Indore.* Rather than treating it as uniformly weak, diagnose what works in Indore and replicate that playbook in its weak markets (Kolkata, Bangalore, Lucknow), supported by festive promotions.

5. Forecast-based planning. At ∼4.8% MAPE, Prophet is accurate enough for weekly operational planning. Maintain baseline stock, pre-plan for the predictable year-end spike, and size safety stock from the forecast's uncertainty band – holding more buffer where it is widest.

6. Data pipeline remediation. The 124-week Kolkata/Furniture gap was a reporting failure, not lost demand; fixing the underlying pipeline prevents similar blind spots from biasing future analysis.

10 Conclusion

This project cleaned, analyzed, and forecast over two years of weekly retail sales for four companies across seven cities and six departments. Two structural anomalies were repaired; exploratory analysis revealed a strong year-end season, regional market leadership, and uneven holiday sensitivity; and the hidden departments were inferred from their demand signatures. Of five forecasting models, Prophet was the most accurate (MAPE = 4.80%), while the more complex SARIMA was the weakest working model. The six-month forecast indicates stable demand with a plannable festive peak, supporting targeted recommendations on inventory, promotions, and regional strategy.

Limitations and future work. The department identities are inferred, not given. Promising extensions include per-department (hierarchical) forecasting reconciled to the company total, adding exogenous regressors (markdowns, weather) via Prophet or SARIMAX, formal rolling cross-validation of the forecast horizon, and learned models (LightGBM, LSTM) as stronger baselines.

References

1. R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, OTexts. https://otexts.com/fpp3/
2. S. J. Taylor and B. Letham, "Forecasting at Scale," *The American Statistician*, 2018.
3. G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Wiley.
4. Statsmodels, pandas, scikit-learn, and Prophet (Python libraries).